Experiment - 9

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D15A/50

Aim - Implement Recurrent Neural Network (RNN) / LSTM for time series or text data.

Dataset Source

Dataset - https://archive.ics.uci.edu/dataset/360/air+quality
Title - Air Quality Dataset

DATASET DESCRIPTION -

The dataset used in this experiment is the Air Quality Dataset obtained from the UCI Machine Learning Repository. It contains hourly averaged responses from an array of chemical sensors deployed in an Italian city to monitor air pollution levels.

The dataset includes several attributes related to air quality measurements such as concentrations of gases and sensor responses. For this experiment, only the **CO(GT)** attribute is used, which represents the concentration of Carbon Monoxide in the air.

The dataset contains missing values represented by -200. These values are replaced with NaN (Not a Number) and removed during preprocessing to ensure data quality. The dataset is then normalized before being used for training the model.

THEORY

1. Time Series Data

Time series data consists of observations recorded sequentially over time. The order of observations is important, as each data point depends on previous values. Unlike traditional datasets where samples are independent, time series data exhibits temporal dependency.

Examples include stock prices, weather data, and pollution levels. In this experiment, the CO concentration values are treated as a time series.

2. Recurrent Neural Networks (RNN)

Recurrent Neural Networks are a class of neural networks designed to process sequential data. They maintain a hidden state that captures information from previous inputs, enabling the network to have memory.

However, standard RNNs suffer from the vanishing gradient problem, which makes it difficult for them to learn long-term dependencies in sequences. As a result, their performance degrades when dealing with long sequences.

3. Long Short-Term Memory (LSTM)

LSTM is an advanced type of RNN designed to overcome the limitations of traditional RNNs. It introduces a memory cell and gating mechanisms that regulate the flow of information.

An LSTM unit consists of three main gates:

- **Forget Gate:** Determines which information from the previous state should be discarded
- **Input Gate:** Decides which new information should be stored in the memory
- **Output Gate:** Controls the output based on the current memory state

These gates enable the LSTM to retain important information over long periods and discard irrelevant data, making it suitable for time-series prediction tasks.

4. Data Preprocessing

The dataset undergoes several preprocessing steps before training:

- Selection of the CO(GT) column as the target variable
- Replacement of missing values (-200) with NaN and removal of such rows
- Normalization of data using MinMaxScaler to bring values between 0 and 1

Normalization helps improve the convergence of the neural network during training.

5. Sequence Generation

Since LSTM requires sequential input, the dataset is converted into input-output pairs using a sliding window approach.

For a time step of 10:

- Input consists of 10 consecutive values
- Output is the next value in the sequence

This allows the model to learn patterns and dependencies across time steps.

6. Model Architecture

The model used in this experiment is a Sequential LSTM model consisting of:

- First LSTM layer with 64 units and return sequences enabled
- Dropout layer to prevent overfitting
- Second LSTM layer with 32 units
- Another dropout layer
- Dense output layer with one neuron for prediction

The model is compiled using the Adam optimizer and Mean Squared Error (MSE) as the loss function.

7. Training the Model

The dataset is split into training and testing sets in an 80:20 ratio. The model is trained using multiple epochs and batch processing.

During training, the model learns to minimize the loss function by adjusting its weights through backpropagation.

8. Prediction

After training, the model is used to predict CO levels on both training and testing datasets. The predicted values are converted back to the original scale using inverse transformation.

9. Evaluation Metrics

Regression Metrics

- **Root Mean Square Error (RMSE):** Measures the standard deviation of prediction errors
- **Mean Absolute Error (MAE):** Measures the average magnitude of errors
- **R^2 Score:** Indicates how well the model explains the variance in the data

Classification Metrics

To evaluate the model from a classification perspective, predicted values are categorized into three levels:

- Low Pollution
- Medium Pollution
- High Pollution

Based on this categorization, the following metrics are computed:

- Accuracy
- Confusion Matrix
- Classification Report (Precision, Recall, F1-score)

10. Hyperparameter Tuning

Various hyperparameters are adjusted to improve model performance:

- Number of LSTM layers
- Number of neurons in each layer
- Dropout rate
- Number of epochs
- Batch size

Proper tuning helps in achieving better accuracy and reducing overfitting.

11. Visualization

Several graphs are used to analyze model performance:

- Actual vs Predicted values plot
- Training and validation loss graph
- Full time-series visualization
- Train-test data split visualization
- Prediction error plot
- Error distribution histogram

These visualizations help in understanding how well the model performs and where it makes errors.

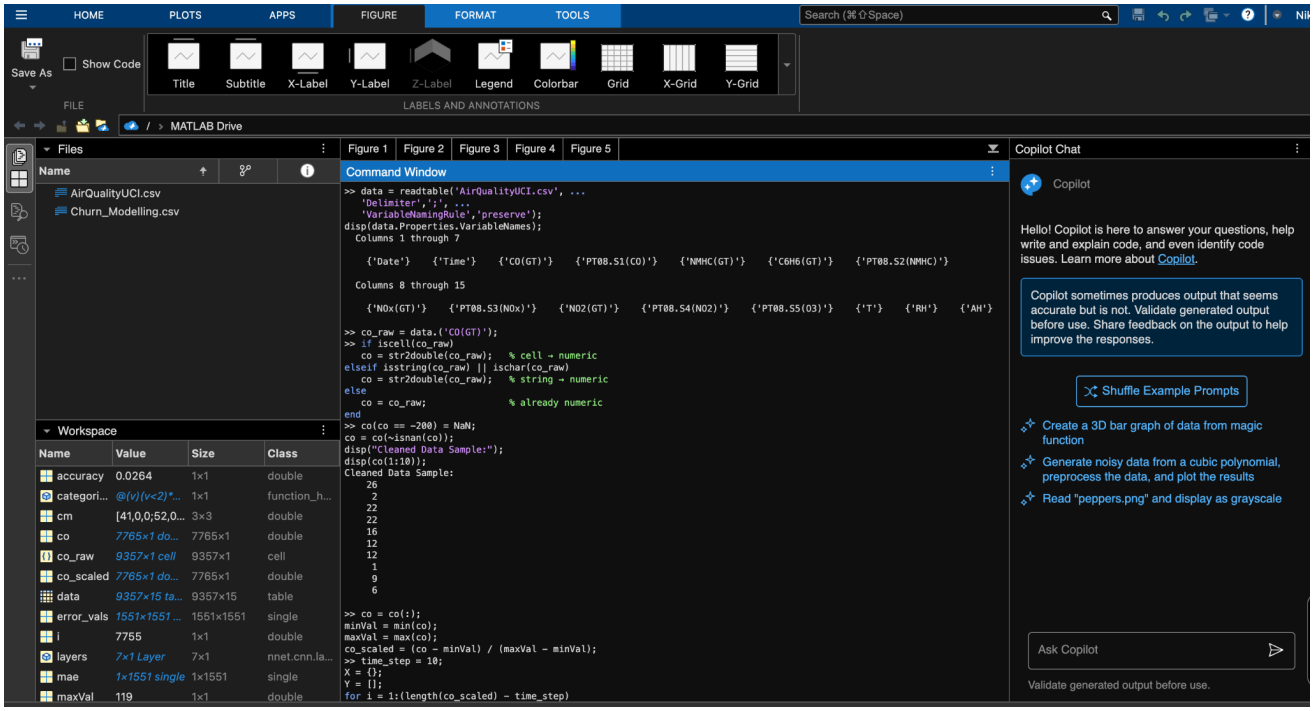
Advantages

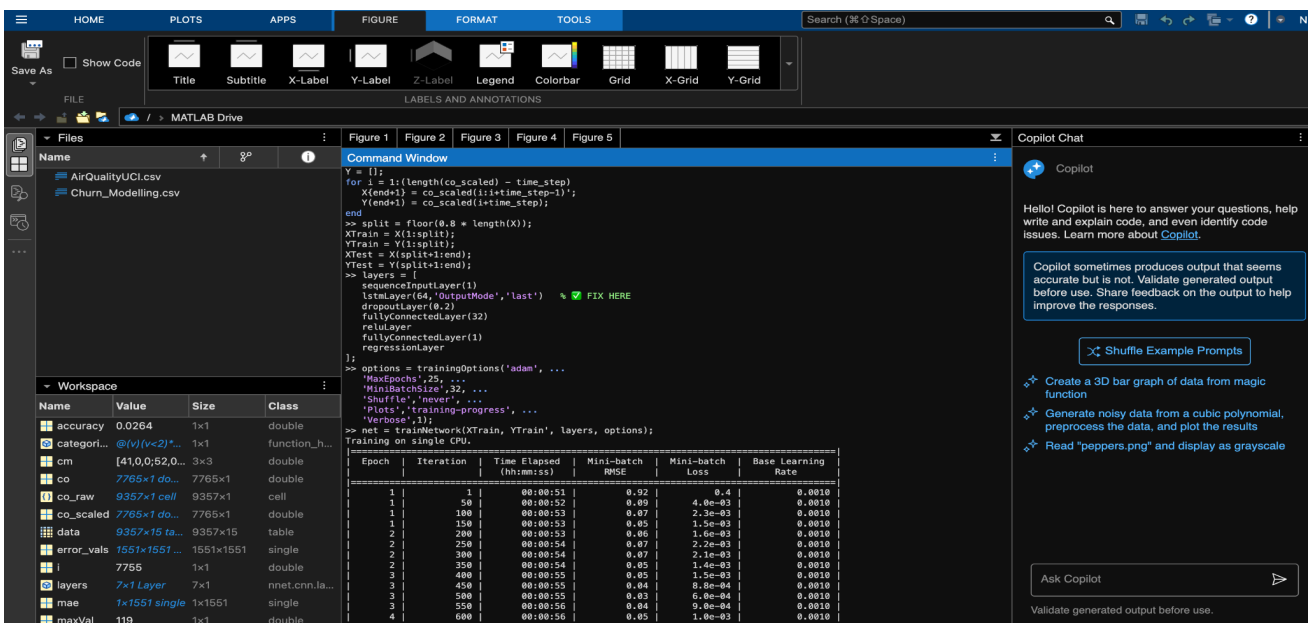
- Effectively handles sequential and time-series data
- Capable of learning long-term dependencies
- Provides better prediction accuracy compared to traditional models
- Useful in real-world forecasting applications

Limitations

- Requires large amounts of data for training
- Computationally expensive and time-consuming

- Complex architecture compared to simple models
- Difficult to interpret internal workings





MATLAB Drive

Files

Name

AirQualityUCI.csv

Churn_Modelling.csv

Figure 1 | Figure 2 | Figure 3 | Figure 4 | Figure 5

Command Window

Explain Error

```
>> YPred = predict(net, XTest);
% Inverse scaling
YPred_actual = YPred * (maxVal - minVal) + minVal;
YTest_actual = YTest * (maxVal - minVal) + minVal;
>> rmse = sqrt(mean((YTest_actual - YPred_actual).^2));
mae = mean(abs(YTest_actual - YPred_actual));
r2 = 1 - sum((YTest_actual - YPred_actual).^2) / ...
         sum((YTest_actual - mean(YTest_actual)).^2);
fprintf('\n--- Regression Metrics ---\n');
fprintf('RMSE: %.4f\n', rmse);
fprintf('MAE: %.4f\n', mae);
fprintf('R2 Score: %.4f\n', r2);

--- Regression Metrics ---
RMSE: 342.4829
RMSE: 344.1966
RMSE: 341.6272
RMSE: 342.4829
RMSE: 1726.9001
RMSE: 338.2124
RMSE: 337.3607
RMSE: 336.5097
RMSE: 337.3607
RMSE: 339.0648
RMSE: 339.9182
RMSE: 339.9182
RMSE: 341.6272
RMSE: 340.7723
RMSE: 340.7723
RMSE: 340.7723
RMSE: 341.6272
RMSE: 341.6272
RMSE: 342.4829
RMSE: 343.3394
RMSE: 346.7729
RMSE: 347.6332
RMSE: 342.4829
RMSE: 344.1966
RMSE: 347.6332
RMSE: 345.9135
RMSE: 339.9182
```

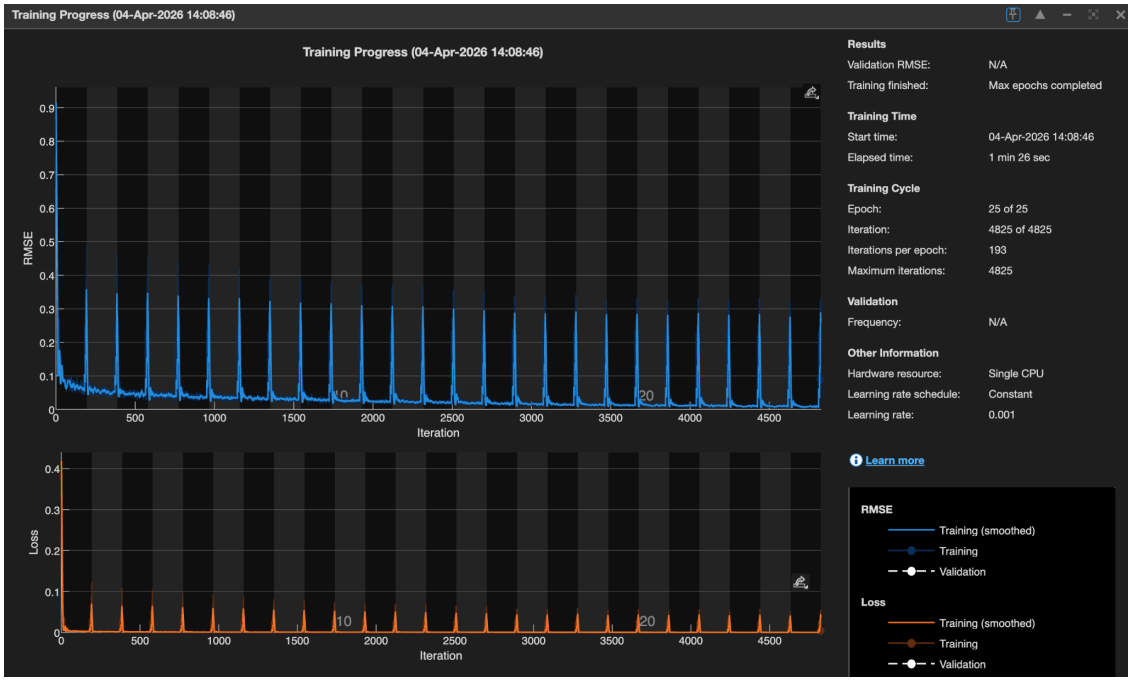
Workspace

Name	Value	Size	Class
accuracy	0.0264	1×1	double
categori...	@(v)(v<2)*...	1×1	function_h...
cm	[41,0,0;52,0...	3×3	double
co	7765×1 do...	7765×1	double
co_raw	9357×1 cell	9357×1	cell
co_scaled	7765×1 do...	7765×1	double
data	9357×15 ta...	9357×15	table
error_vals	1551×1551 ...	1551×1551	single
i	7755	1×1	double
layers	7×1 Layer	7×1	nnet.cnn.la...
mae	1×1551 single	1×1551	single
maxVal	119	1×1	double

```
R2 Score: -0.4689
R2 Score: -0.4474
R2 Score: -0.4545
>> categorize = @(v) (v<2)*0 + (v>=2 & v<5)*1 + (v>=5)*2;
y_test_cat = arrayfun(categorize, YTest_actual);
y_pred_cat = arrayfun(categorize, YPred_actual);
cm = confusionmat(y_test_cat, y_pred_cat);
accuracy = sum(diag(cm)) / sum(cm(:));
fprintf('\n--- Classification Metrics ---\n');
fprintf('Accuracy: %.4f\n', accuracy);
disp("Confusion Matrix:");
disp(cm);

--- Classification Metrics ---
Accuracy: 0.0264
Confusion Matrix:
          41           0           0
          52           0           0
        1458           0           0

>> figure;
heatmap(cm);
title('Confusion Matrix');
figure;
plot(YTest_actual,'b'); hold on;
plot(YPred_actual,'r');
legend('Actual','Predicted');
title('Actual vs Predicted');
figure;
plot(co);
title('Full Time Series');
figure;
error_vals = YTest_actual - YPred_actual;
plot(error_vals);
title('Prediction Error');
figure;
histogram(error_vals,50);
title('Error Distribution');
```

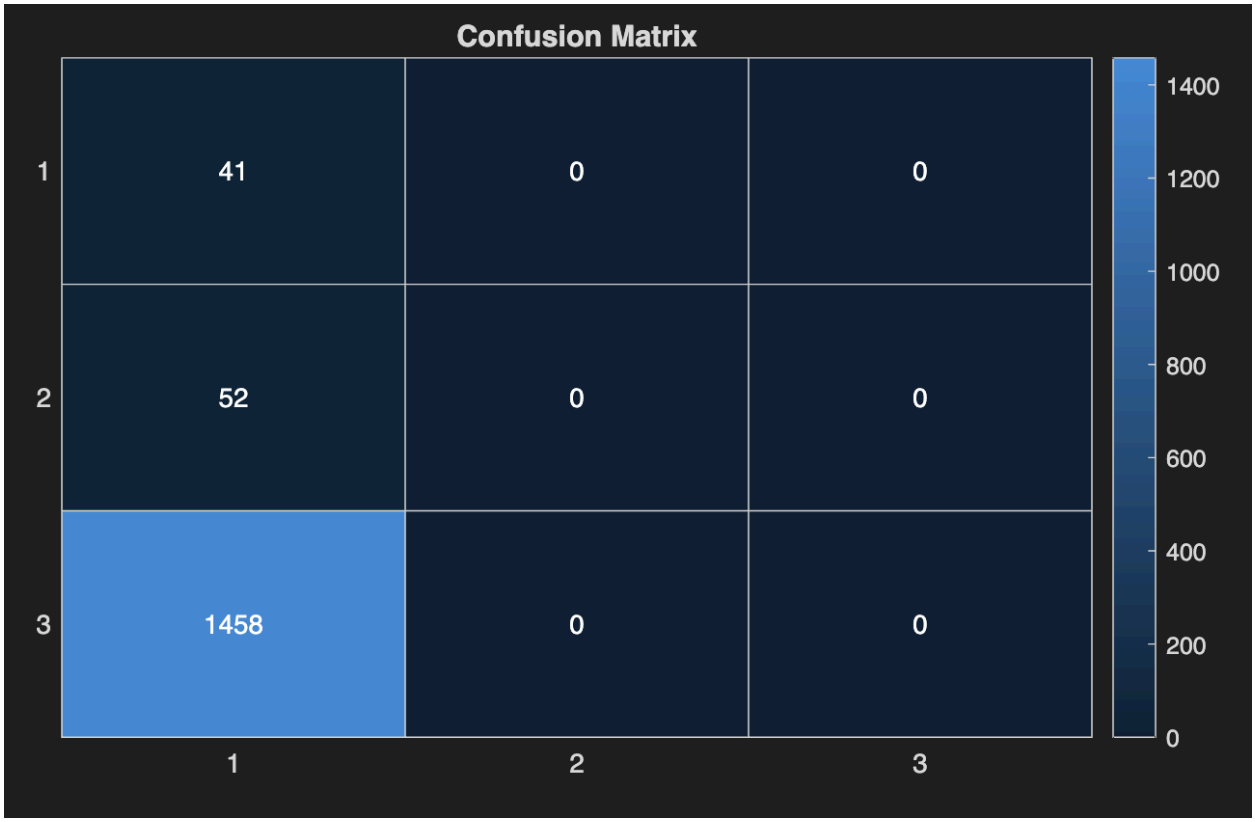


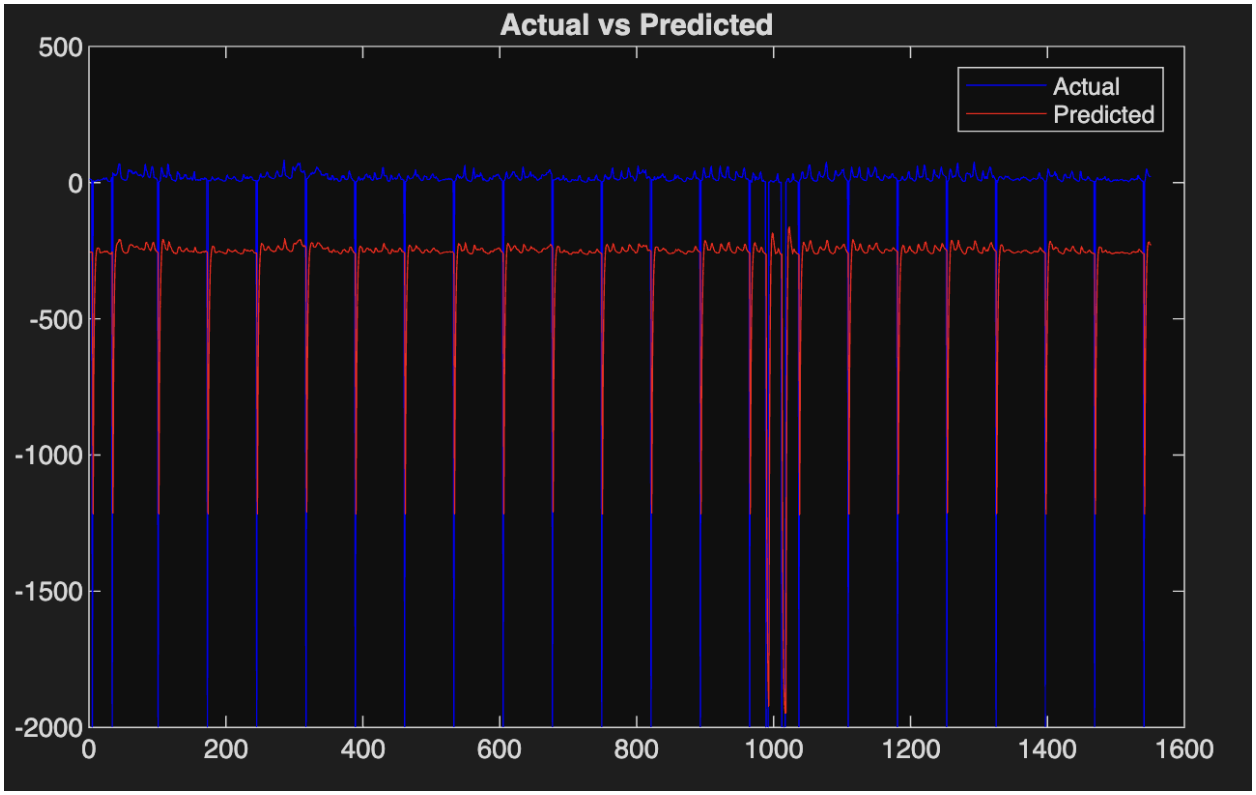
Command Window

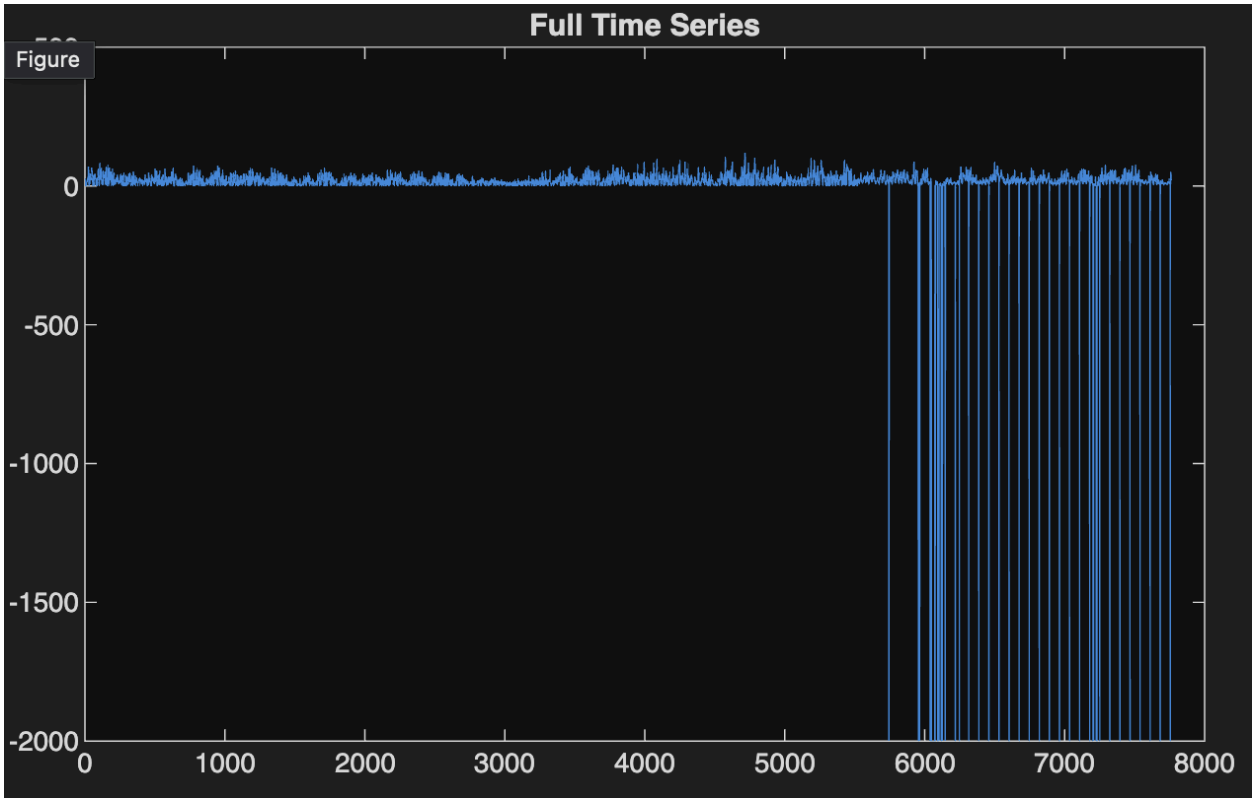
15	2800	00:01:12	0.02	1.9e-04	0.0010
15	2850	00:01:12	0.02	3.0e-04	0.0010
16	2900	00:01:12	0.02	2.6e-04	0.0010
16	2950	00:01:13	0.02	1.6e-04	0.0010
16	3000	00:01:13	0.02	1.4e-04	0.0010
16	3050	00:01:13	0.02	1.2e-04	0.0010
17	3100	00:01:14	0.04	6.3e-04	0.0010
17	3150	00:01:14	0.02	1.5e-04	0.0010
17	3200	00:01:14	0.02	1.1e-04	0.0010
17	3250	00:01:15	0.02	1.7e-04	0.0010
18	3300	00:01:15	0.03	5.4e-04	0.0010
18	3350	00:01:15	0.02	1.5e-04	0.0010
18	3400	00:01:16	0.02	1.5e-04	0.0010
18	3450	00:01:16	0.01	9.2e-05	0.0010
19	3500	00:01:16	0.03	3.4e-04	0.0010
19	3550	00:01:17	0.01	7.7e-05	0.0010
19	3600	00:01:17	0.01	8.2e-05	0.0010
19	3650	00:01:17	0.01	8.6e-05	0.0010
20	3700	00:01:18	0.02	1.6e-04	0.0010
20	3750	00:01:18	0.01	5.9e-05	0.0010
20	3800	00:01:19	0.02	1.4e-04	0.0010
20	3850	00:01:19	0.01	7.1e-05	0.0010
21	3900	00:01:19	0.01	7.1e-05	0.0010
21	3950	00:01:20	9.08e-03	4.1e-05	0.0010
21	4000	00:01:20	0.01	6.1e-05	0.0010
21	4050	00:01:20	0.28	3.9e-02	0.0010
22	4100	00:01:21	0.01	7.2e-05	0.0010
22	4150	00:01:21	0.01	7.9e-05	0.0010
22	4200	00:01:21	0.01	8.1e-05	0.0010
23	4250	00:01:22	0.04	8.9e-04	0.0010
23	4300	00:01:22	0.01	6.4e-05	0.0010
23	4350	00:01:22	9.07e-03	4.1e-05	0.0010
23	4400	00:01:23	0.01	8.1e-05	0.0010
24	4450	00:01:23	0.04	7.3e-04	0.0010
24	4500	00:01:23	0.01	7.0e-05	0.0010
24	4550	00:01:24	7.65e-03	2.9e-05	0.0010
24	4600	00:01:24	9.67e-03	4.7e-05	0.0010
25	4650	00:01:24	0.03	5.1e-04	0.0010
25	4700	00:01:25	0.01	7.4e-05	0.0010
25	4750	00:01:25	8.79e-03	3.9e-05	0.0010
25	4800	00:01:25	8.52e-03	3.6e-05	0.0010
25	4825	00:01:26	0.09	3.9e-03	0.0010

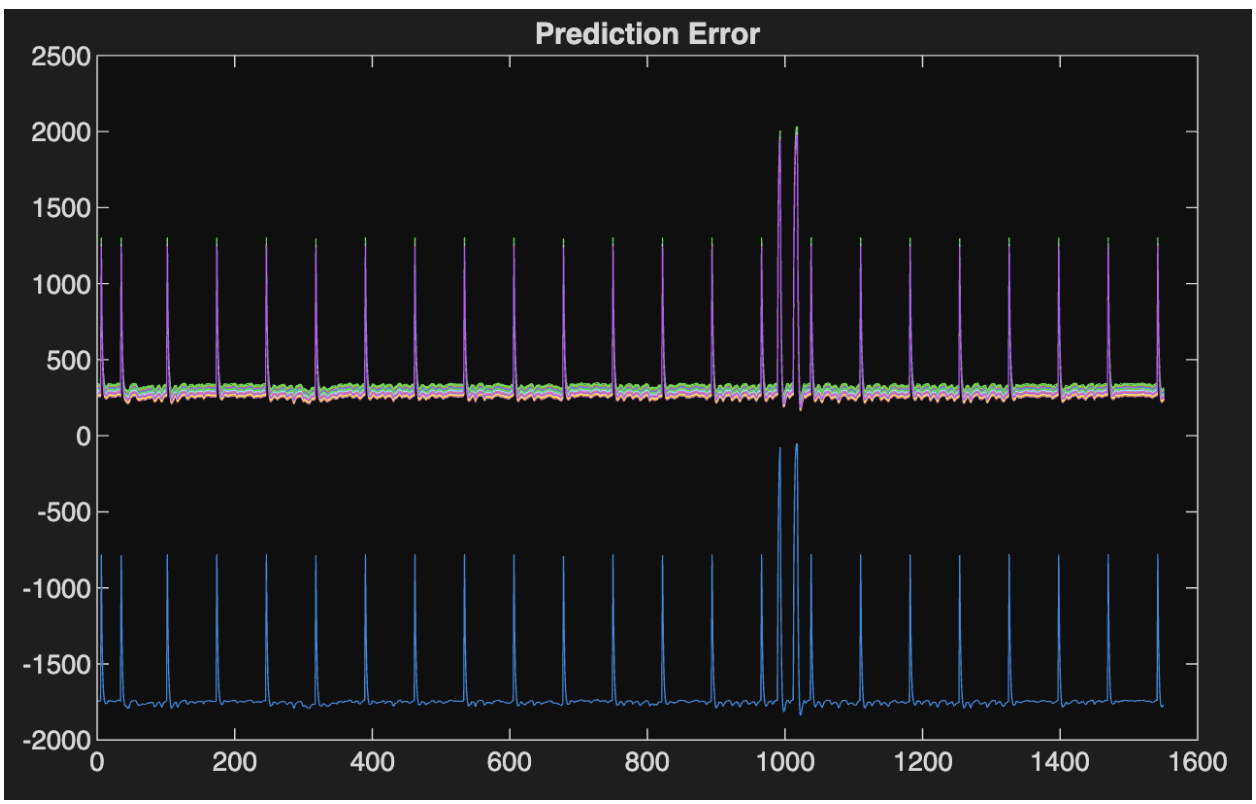
Training finished: Max epochs completed.

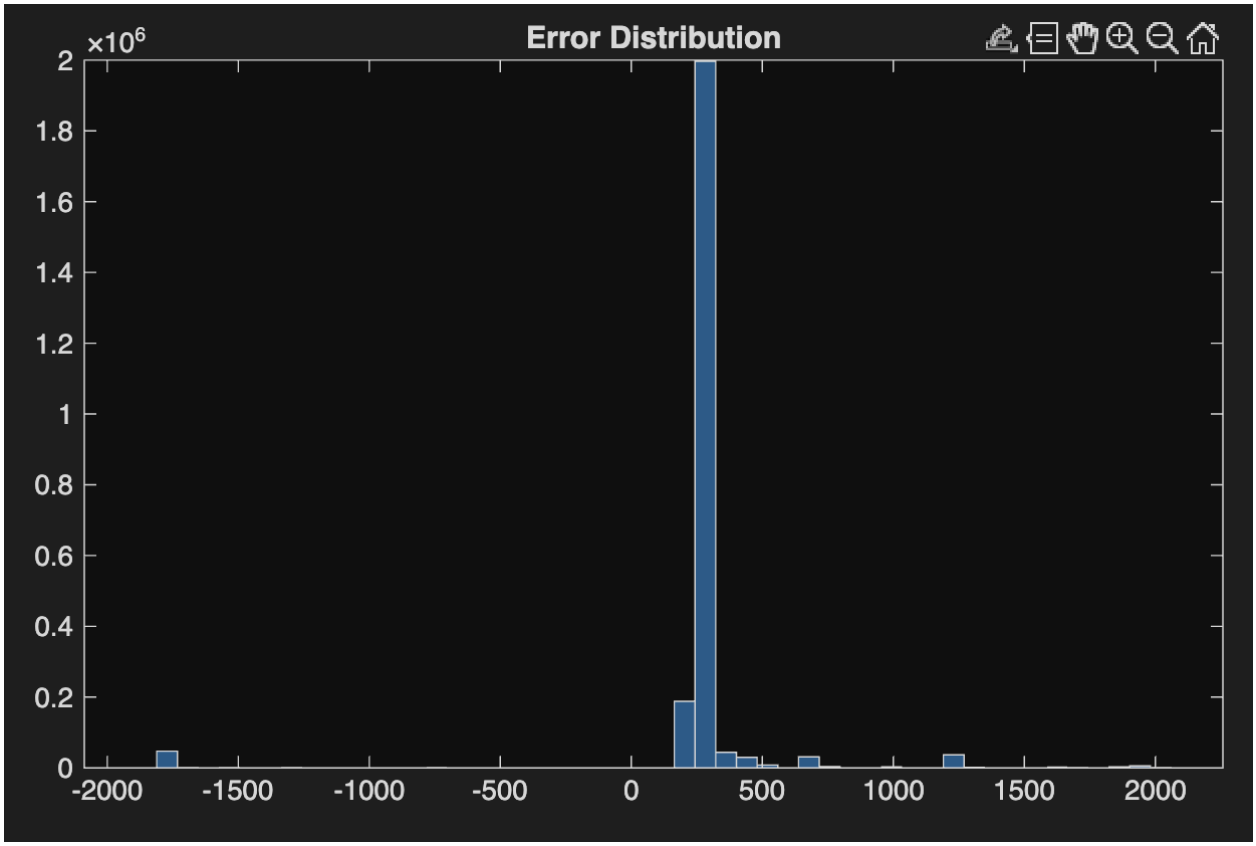
>>


Confusion Matrix
1
2
3
41
0
0
52
0
0
1458
0
0
1
2
3
1400
1200
1000
800
600
400
200
0


Actual vs Predicted
Actual
Predicted
500
0
-500
-1000
-1500
-2000
0
200
400
600
800
1000
1200
1400
1600


Full Time Series
Figure
0
-500
-1000
-1500
-2000
0
1000
2000
3000
4000
5000
6000
7000
8000


Prediction Error
2500
2000
1500
1000
500
0
-500
-1000
-1500
-2000
0
200
400
600
800
1000
1200
1400
1600



CONCLUSION

In this experiment, an LSTM-based neural network was implemented to perform time-series prediction on air quality data. The model successfully learned patterns from historical CO values and predicted future values with reasonable accuracy.

The preprocessing steps ensured that the dataset was clean and normalized, which contributed to stable training. The sequence generation process enabled the model to capture temporal dependencies effectively.

The model was evaluated using both regression and classification metrics, providing a comprehensive understanding of its performance. Visualization techniques further helped in analyzing trends, errors, and training behavior.

Overall, the experiment demonstrates the effectiveness of LSTM networks in handling time-series data and highlights their applicability in environmental monitoring and forecasting systems.